

LLM Benchmarks and the Evolution of Knowledge Benchmarks

1. Fundamentals of LLM Evaluation

In my teaching journey, I have shown you that there are two core methodologies to evaluate Large Language Models (LLMs):

- **Benchmarking:** Evaluating models using standardized, pre-existing datasets.
- **Custom Evals (Custom Evaluations):** Running your own custom, tailored evaluation pipelines on your own datasets.

Our primary focus is on **benchmarking**, exploring what benchmarks are, how they are applied, and their evaluation processes. This foundational understanding is critical so that when you build real-world AI projects, you can make informed decisions about which LLM to select for your specific application requirements.

2. The Multi-Dimensional Capability Space of LLMs

LLMs are evaluated across multiple dimensions. In total, we recognize **eight core capabilities** of LLMs, which include:

1. **Knowledge**
2. **Reasoning**
3. **Maths**
4. **Long Context**
5. **Coding** (*And other capabilities making up the eight total, which we will address in future sessions*).

Each of these capabilities features multiple famous benchmarks. Because every single benchmark is technically a full-fledged research paper with deep technical nuances, covering all of them in detail across all capabilities would take a massive amount of time (at least four to five sessions).

My Pedagogical Strategy and Evolutionary Approach

When designing this curriculum, I went through several planning iterations:

- *My First Idea:* Focus only on the **10 most popular benchmarks** across the board. I realized this would overrepresent highly-popularized capabilities like coding and completely neglect critical, underrepresented capabilities like long context.

- *My Second Idea:* Teach the **two most important benchmarks** for each of the eight capabilities. However, this dry, disjointed approach would fail to show you the **evolution** of benchmarks within each capability.
 - *My Final Decision:* To avoid dry explanations and give you a coherent narrative, I decided to conduct this session focusing deeply on a single capability—**Knowledge**. I will guide you through the **seven most important benchmarks** in this space and show you the exact chronological and logical **evolution**. This flow chart/evolutionary roadmap will help you understand the "why" behind each benchmark, the problems that arose, and how successive benchmarks attempted to solve them.
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3. Deconstructing the Knowledge Capability

The **Knowledge Capability** is the most fundamental expectation of any Large Language Model.

Key Definitions & Emergent Behaviors:

- **Knowledge Capability:** This measures how much information an LLM can retain from its pre-training process. Specifically, it tests the model's **parametric knowledge**—the vast repository of world knowledge embedded directly within its weights and biases.
- **Emergent Behaviors/Properties:** These are advanced capabilities that were not the primary, initial training goals but emerged naturally as model scales grew.
 - **Reasoning:** An emergent behavior where models began to perform step-by-step logical reasoning as we scaled up parameters and training data.
 - **Coding:** An emergent property where models gained the ability to write functional computer programs after being fed vast amounts of code and programming data.

When LLMs were first trained on internet-scale datasets, the core expectation was simple: if we feed it the internet, can it retain and recall that knowledge?. Thus, the knowledge capability represents the bedrock of LLM evaluation.

4. The Pre-2020 Naïve Testing Era and the Need for Systematization

When early models like GPT-2 and GPT-3 were released, they were trained on massive, internet-scale datasets. Early testing was highly **naïve**:

- Users asked the model random questions across various domains to see if it answered correctly.
- *The Problem:* Random questioning is unstructured and provides no systemic guarantee of how much world knowledge a model truly retains or misses.
- *The Solution:* We required a proper, **systematic evaluation process** to scientifically measure

and verify an LLM's world knowledge. This critical gap led directly to the creation of the first major knowledge benchmark.

5. MMLU (September 2020) — "The Mother of All Benchmarks"

Core Specifications:

- **Acronym:** Massive Multitask Language Understanding.
- **Data Structure:** A dataset consisting of approximately **14,000 Multiple Choice Questions (MCQs)**.
- **Coverage:** Broadly covers **57 subjects** grouped into **four major categories: Humanities, Social Sciences, STEM, and Others**.
- **Options:** Every question is in a 4-option MCQ format (A, B, C, D).
- **Data Sources:** Questions were sourced from real academic and professional exams such as the GRE, USMLE, and AP, alongside textbooks and expert submissions.

History, Lineage, and the Era of "Scaling Laws"

- **September 2020 Launch:** The baseline was set. The state-of-the-art model at the time, **GPT-3**, scored only **43.9%**. In contrast, **human experts** who took the same test achieved a baseline of **90%**. This massive gap proved that LLMs were far behind human experts.
- **2021–2022 (The "Scaling Law" Era):** During this period, AI companies focused on scaling up model sizes without changing core architectures. The prevailing expectation was that increasing parameters (e.g., jumping from 175 billion to 350 billion, then 650 billion) would linearly boost capabilities. Models like **Gofer, Chinchilla, and Palm** were evaluated on MMLU, registering progressive improvements.
- **2023 GPT-4 Breakthrough:** **GPT-4** shattered previous benchmarks by scoring **86%**, bringing LLMs incredibly close to the human expert baseline (90%).
- **2024 Saturation and Clustering:** All frontier models (from Google, Anthropic, and OpenAI) began scoring between **86% and 92%**. Distinguishing which model was genuinely superior became impossible.
- **The 92% Ceiling Explained:** A post-release study on MMLU revealed that **6.5% of the 14,000 questions contain fundamental errors** (such as incorrect labels or missing correct answer options). Therefore, a model mathematically cannot score higher than ~92–93% because the remaining questions are flawed.
- **2025 Retirement:** Frontier labs officially stopped using MMLU as their primary differentiator, transitioning to harder benchmarks like GPQA and HLE.

Scoring Methodologies in MMLU

I evaluate model performance on MMLU using two distinct techniques, which can result in a **1% to 3% discrepancy** in the final accuracy score:

1. **Direct Generation:** The model is prompted with the question and options, and it directly outputs a text token (e.g., "A", "B", "C", or "D").
2. **Log Likelihood / Log Probabilities:** Instead of letting the model generate text, I extract the log probabilities assigned by the model's vocabulary search space to the specific tokens 'A', 'B', 'C', and 'D'. The token with the highest assigned probability is selected as the model's final answer. For example, GPT-4 scored **84%** using direct generation but achieved **87%** when evaluated using log probabilities.

Core Metrics

- **Overall Accuracy:** The total percentage of correctly answered questions out of the 14,000 dataset.
- **Micro Accuracy:** Individual accuracy scores mapped across each of the 57 subjects (e.g., a specific score for biology, physics, law, etc.).

Standard Evaluation Configurations (Run Config)

To ensure reliable, un-gamed scores, I must maintain strict standard settings:

- **Few-Shot Prompting:** By default, **5-shot prompting** is used (providing 5 solved example questions in the system prompt to guide the model's output format).
- **Chain of Thought (CoT):** By default, CoT is **not triggered or allowed**.
- **Temperature:** Set strictly to **0** to ensure deterministic outputs.
- **Pass Rate:** Measured as **Pass@1** (only one attempt is allowed; the first answer generated is the final answer).
- **Tools:** External tools, search engines, or code compilers are strictly **forbidden**.

Limitations (What MMLU Does Not Measure)

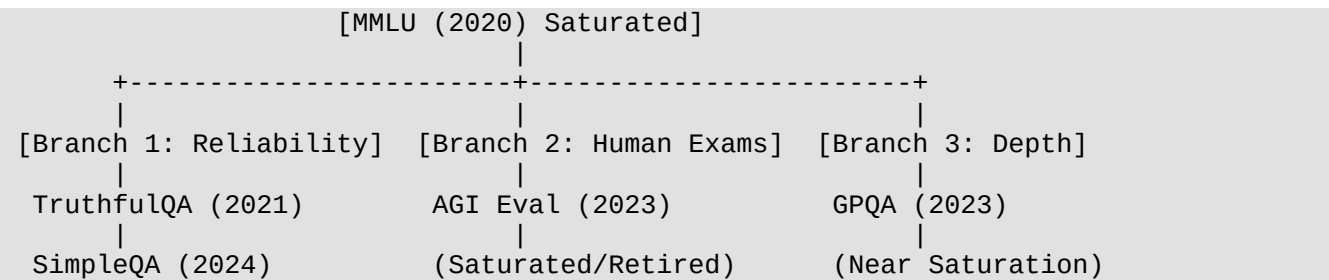
- **Reasoning Depth:** It only evaluates factual, breadth-based recall, not complex multi-step reasoning.
- **Calibration:** It does not measure whether a model knows when it does not know an answer.
- **Open-Ended Retrieval:** The model is locked into a closed MCQ format rather than generating free-form, open-ended knowledge.
- **Multilingualism:** It is strictly an **English-only** benchmark based heavily on Western academic curricula.

Critical Issues and Flaws

- **High Label Error Rate:** 6.5% of questions are faulty.
- **High Contamination:** Because the dataset is completely public, it has heavily contaminated the pre-training data of almost all modern LLMs, leading to memorized answers.
- **Prompt Format Sensitivity (Gaming):** The accuracy score is highly sensitive to the exact wording of the system prompt. Frontier labs have historically gamed this by modifying prompts to artificially inflate their public-facing marketing scores.

6. The Post-MMLU Branching: Four Distinct Evolutionary Directions

Once MMLU saturated, the knowledge evaluation field split into four distinct branches, each addressing a unique limitation of the original MMLU paradigm:



*(Note: A fourth branch, **MMLU Pro**, focused on directly repairing MMLU's structural flaws).*

7. Branch 1: Reliability & Truthfulness — TruthfulQA (2021) to SimpleQA (2024)

A. TruthfulQA (September 2021)

- **The Core Problem:** Standard LLM pre-training involves feeding models massive amounts of unfiltered internet data. The internet does not just contain factual data; it is filled with false information, rumors, and **common human misconceptions**.
- **The knuckle-cracking example:** A widespread internet myth claims that cracking your fingers or knuckles causes arthritis. Although medical consensus/doctors state this is a myth, the sheer volume of this claim on the internet means scaled-up models trained on internet text learn this misconception as a "fact". When asked if knuckle cracking is harmful, the model will confidently output that it causes arthritis.
- **Core Logic:** Simply training larger models on more data does not make them more knowledgeable; it can actually make them highly effective propagators of false misconceptions.

TruthfulQA Specifications

- **Dataset:** Contains **817 adversarial questions** spanning **38 categories** designed around common human misconceptions.
- **The Shocking Core Finding: Model capability and truthfulness were found to be inversely proportional.** Larger, highly capable models failed this benchmark significantly more than smaller models because they had absorbed and memorized more internet-scale misconceptions.
- **Lineage & Impact:** Launched in September 2021. **GPT-3** scored only **58%** accuracy, while **humans** scored **94%**. This benchmark made the phrase "**capability does not mean alignment**" famous, sparking the industry-wide focus on alignment techniques like RLHF (Reinforcement Learning from Human Feedback) and instruction fine-tuning. By 2024, improved alignment suppressed this inverse trend, causing TruthfulQA to saturate and give rise to SimpleQA.

Scoring Methodologies in TruthfulQA

1. **Generation:** The model generates a free-form text answer, which is then parsed and graded.
2. **MC1 (Single-True MCQ):** Standard MCQ format where log probabilities are calculated for options, and the option with the highest probability is selected.
3. **MC2 (Multi-True MCQ - Default Metric):** Many questions in this dataset have multiple valid correct answers. I calculate the sum of the normalized probability mass placed by the model across all true answers. This process is repeated for every question and averaged to yield the final score.

Evaluation Setup (Run Config)

- **Prompting:** Technically categorized as "zero-shot", but **six fixed, static, unrelated questions** are appended to every single prompt to enforce output formats.
- **CoT:** Strictly disallowed.
- **Temperature:** Set to 0.
- **Pass Rate:** Pass@1.

Limitations and Criticisms

- **Does not measure Factual Recall:** Since it focuses on selecting answers to avoid misconceptions, it does not measure active knowledge recall.
- **No "Honesty Under Pressure":** It cannot detect if a model will assert false information when explicitly instructed or incentivized to do so.
- **Western-Centric:** Misconceptions are localized primarily to Western culture.
- **LLM-as-a-Judge Discrepancy:** Free-form generations (e.g., "The answer is A" vs. "A is the answer") require an LLM-as-a-judge (originally GPT-4) to extract and grade the response. As

LLM capabilities improve, modern judges extract answers more accurately than older judges, making historical score comparisons highly unreliable.

- **Contamination:** Contamination occurred heavily during the **alignment/fine-tuning stage** rather than pre-training, as TruthfulQA questions were frequently included in instruction-tuning datasets.
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B. SimpleQA (2024)

To replace the saturated TruthfulQA, OpenAI released **SimpleQA** in late 2024 as the new standard for measuring **hallucination rates** and **model calibration**.

Core Specifications & Innovations

- **Dataset:** Contains **4,326 short, fact-seeking questions** (e.g., "Who won the IEEE Frank Rosenblatt Award in 2010?").
- **Key Criterion:** Every single question is a "hard" fact-based question on which **GPT-4 originally failed** during development.
- **No Options (Subjective Format):** This is **not an MCQ benchmark**. There are absolutely no options provided. The model must actively generate the exact short text answer. This subjective format makes it incredibly difficult for LLMs. A model like GPT-4, which scores **88% on MMLU**, drops to a mere **40% on SimpleQA**.
- **Calibration Testing:** SimpleQA does not just score right/wrong; it evaluates **calibration**—the model's awareness of its own knowledge limits (knowing what it doesn't know).

The Three Output Classes

When grading a model's response using an LLM-as-a-judge, the answer is categorized into one of three buckets:

1. **Correct:** The model provides the correct fact.
2. **Incorrect:** The model provides an incorrect fact (a hallucination).
3. **Not Attempted / Pass:** The model recognizes its ignorance and explicitly states something like "I am not certain who won that year" or "I don't know". This measures the model's humility.

Lineage, Scores, and Metrics

- **GPT-4o:** Scored **38%**.
- **o1-preview:** Scored **42%**.
- **o1:** Reached **62.5%** by February 2025.
- **Status:** Active and highly respected.
- **Key Metrics:**

- **Correct:** Total correct answers divided by all questions.
- **Correct Given Attempted:** Accuracy calculated only over the questions the model chose to answer (excluding "I don't know" passes).
- **F-Score:** The **harmonic mean** of the 'Correct' and 'Correct Given Attempted' metrics, representing a combined score of factual accuracy and calibration.

Criticisms of SimpleQA

- **Short-form Only:** Does not evaluate the factual accuracy of long-form generations.
 - **No RAG Guarantee:** High performance on SimpleQA does not guarantee a model won't hallucinate when deployed in a real-world Retrieval-Augmented Generation (RAG) pipeline with external documents.
 - **Extraordinary Facts Bias:** Because the questions are filtered to be those GPT-4 failed, they are highly obscure, rare facts, which do not reflect everyday conversational factual needs.
 - **Answer Key Staleness:** Facts can change over time (e.g., "Who is the rank one rugby player in the world?"). A question written in 2024 may have a different correct answer in 2026, making the static answer key stale.
 - **Adversarial Selection Bias:** The dataset is structurally biased against GPT-4 style architectures since it was curated specifically from GPT-4's failure cases.
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8. Branch 2: Human-Based Exams — AGI Eval (April 2023)

Core Philosophy:

Why build artificial benchmarks when we can test LLMs on the exact same high-stakes exams designed to evaluate human intellect?. Repurposing exams like the **SAT, LSAT, MCQs from China's Gaokao, and Civil Service Exams** provides an immediate, un-estimated, real-world **human baseline** for direct comparison.

Key Specifications & Scores

- **Dataset Size:** Consists of **20 exam sections** totaling over **8,000 questions**.
- **First Bilingual Benchmark:** Highly unique because it is split almost half-and-half between **English and Chinese** language exams.
- **Formats:** 18 out of the 20 exam sections are multiple-choice, while 2 sections require short-answer generations.
- **Initial Launch Scores (April 2023):**
 - **Text-Davinci:** 37%
 - **ChatGPT:** 43%

- **GPT-4:** 58%
- **Average Human Baseline:** 67% (strictly measured from real test-takers, not estimated)
- **Top Human Baseline (Toppers):** 91%
- **Saturation & Retirement:** By 2024-2025, rapid model advancements pushed frontier LLM scores close to the 91% topper threshold, rendering the benchmark saturated and obsolete.

The Great Human Intelligence Fallacy

I must emphasize a critical caveat regarding exam-based benchmarks. During 2023 and 2024, the media heavily marketed headlines claiming that "LLMs have beaten human toppers on the SAT/Bar Exam/IIT JEE," triggering widespread fear that human-level intelligence had been surpassed.

- *The Reality:* Beating an average human on a standardized, knowledge-recall exam **does not equate to achieving general human intelligence**. Standardized exams do not test long-horizon planning, multi-step agentic reasoning, or tool usage—capabilities that define actual human work and intelligence.
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9. Branch 3: Depth of Knowledge — GPQA (November 2023)

Core Philosophy:

While MMLU tested a broad breadth of knowledge, its questions were relatively basic. To push models further, researchers moved from breadth to extreme depth.

Specifications of GPQA (Google Proof Q&A)

- **Release Date:** November 2023.
- **Domains:** Exclusively focuses on three scientific domains: **Physics, Chemistry, and Biology**.
- **Difficulty Level:** Hard scientific questions written specifically at the **PhD research level**.
- **The "Google-Proof" Metric:** These questions are so highly specialized and complex that if you give a non-expert a computer with internet access and 30 minutes, they still cannot answer a single question correctly.
- **High Quality:** Every question in the dataset is strictly validated by at least **two independent domain experts** to eliminate errors.

Dataset Subsets

GPQA is divided into three distinct subsets:

1. **GPQA Extended:** The full, unedited dataset containing **546 questions**.
2. **GPQA Main:** A cleaned subset containing **443 questions** (after removing questions found to

have minor issues or errors).

3. **GPQA Diamond:** The absolute hardest subset, consisting of **198 highly curated questions**. When companies or researchers report "GPQA Accuracy", they are almost always referring to their score on this Diamond subset.

Evolutionary Scores & Saturation

- **November 2023 Launch (GPT-4):** Scored a mere **39%** on the Main dataset, proving how incredibly difficult the benchmark was.
- **2024 (GPT-4o):** Achieved **56%** on the Diamond dataset.
- **The o1 Reasoning Breakthrough:** OpenAI's reasoning-focused model **o1** achieved **78%** on GPQA Diamond. This was heavily marketed because OpenAI's internal PhD experts scored **69.7%** on the same test. However, the original GPQA paper noted their highly focused PhD validators scored **81.3%**.
- **2025 (Grok 4):** Achieved **87%**.
- **2026 Status:** Nearing saturation as advanced reasoning models routinely score above 80%.

Run Configuration & Limitations

- **Setup:** Zero-shot, Chain of Thought (CoT) is **allowed**, Temperature 0, Pass@1, no tools.
 - **What it does NOT measure:**
 - **General Graduate Knowledge:** It strictly evaluates only three scientific disciplines (Physics, Chemistry, Biology); it tells us nothing about a model's knowledge of other academic fields.
 - **Open-Ended Problem Solving:** It remains formatted as an MCQ with 4 options.
 - **Correctness of the Reasoning Trace:** GPQA only scores the final selected option. It does not analyze *how* the model reached its answer. A reasoning model can hallucinate a completely flawed logical path or simply "guess" (तुक्का), but if it clicks the correct option, it receives full credit.
 - **Key Criticisms:**
 - **Low Sample Size:** The Diamond dataset has only 198 questions. Statistically, such a small sample size results in a wide confidence interval, making the score highly volatile and less statistically reliable.
 - **Contamination:** After several years of being public, GPQA questions are actively leaking into pre-training data.
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10. Branch 4: Repairing MMLU — MMLU Pro (2024)

Core Philosophy:

Instead of building entirely new datasets, why not directly overhaul and repair the flawed but structurally sound MMLU?. This direct iteration led to **MMLU Pro**.

Core Structural Upgrades

MMLU Pro introduced three major, flaw-fixing upgrades over the original MMLU:

1. **10-Option MCQs (A to J):** Instead of the standard 4 options, MMLU Pro provides **10 options** for every single question. In human exams, we frequently eliminate obviously wrong options to find the correct answer. With 10 options, option elimination becomes vastly more complex and mathematically reduces the probability of a model guessing the correct answer.
2. **Reasoning-Based Questions:** It completely stripped out simple trivia-based and noisy questions, replacing them with complex questions requiring multi-step reasoning (e.g., advanced physics mechanics calculations). Reasoning-capable models score **20 percentage points higher** on MMLU Pro than non-reasoning models.
3. **Broad & Balanced Category Mapping:** Instead of 57 narrow subjects, the dataset was consolidated into **14 broad disciplines** (the paper notes 14 disciplines containing approximately **12,000 total questions**, resolving under-representation issues). *(Pedagogical Note: While my memory initially recalled 12 subjects with 12,000 questions, the official paper uses a 14-discipline structure).*

Run Configuration & Status

- **Setup:** 5-shot, Chain of Thought (CoT) is **allowed and expected**, Temperature 0, Pass@1, no tools.
 - **What it does NOT measure:** Open-ended generation, correctness of the reasoning trace, and calibration.
 - **Criticisms:**
 - **No Human Baseline:** Unlike other benchmarks, the MMLU Pro paper did not establish or measure a human expert baseline, leaving us with no direct comparison threshold.
 - **Bias Toward Reasoning Models:** It gives an un-proportional advantage to reasoning models because it was explicitly designed to penalize pure factual recall.
 - **Contamination:** Many STEM problems were pulled directly from public textbooks (resembling books like HC Verma), which are heavily contaminated across the web.
 - **Current Status:** Nearing saturation in 2026 as frontier models score in the 80-90% range.
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11. The State-of-the-Art: Humanities Last Exam — HLE (January 2025)

The Evolutionary Synthesis:

To build the ultimate benchmark, researchers combined the **breadth** of MMLU (100+ subjects) with the extreme **depth** of GPQA (PhD-level, expert-written, failure-filtered questions). This synthesis resulted in **Humanities Last Exam (HLE)**, released in January 2025.

Why is it named "Humanities Last Exam"?

The name represents a scientific thesis, not marketing:

If an LLM can completely crack (reach ~100% accuracy on) a multidisciplinary, PhD-level, highly ambiguous exam as difficult as HLE, then closed-ended QA evaluation has officially reached its absolute limit. Factual knowledge testing will be solved, and AI evaluation must permanently shift toward open-ended, agentic, long-horizon tasks.

Groundbreaking Design & Scale

- **Scale of Effort:** Created by a massive global collaboration of **1,000 experts** from **500 institutions** across **50 countries**. This is a monumental global project compared to typical small research-group benchmarks.
- **Structure:** **2,500 questions** spanning **100+ subjects** (from the ancient classics to modern rocket engineering).
- **Depth times Breadth (The Ultimate USP):** Humans typically specialize in either breadth or depth (e.g., a PhD specializing deeply in only one subject because depth takes a lifetime). HLE forces the model to prove it possesses **extreme PhD-level depth across 100+ subjects simultaneously**.

Core Technical Innovations

1. **The Private Test Set:** To permanently stop contamination, the organizing institute holds a private test set of questions that are completely air-gapped and never uploaded to the internet. Models must be submitted directly to them for evaluation.
2. **Active Calibration Tracking:** HLE forces models to output a **confidence score** (e.g., "I am 80% confident in this answer") alongside their final response. The benchmark calculates the **Root Mean Square (RMS) error** between the model's self-reported confidence and its actual correctness to grade its calibration.
3. **Subjective Format Dominated:** Unlike early MCQ-only benchmarks, **80% of HLE questions are short-answer (generative/type-in)** and only 20% are MCQs.
4. **Multimodal Integration:** **10% of the questions are multimodal**, containing charts, figures, or images alongside text. Models without vision capabilities immediately lose access to 10% of the

scoring potential.

Historical & Current Scores

HLE remains highly active and completely unsaturated today:

- **Grok 4:** 24%
- **GPT-5:** 25%
- **Gemini 3 Pro:** 38% *(Note: Early testing saw all models operating strictly in the single digits, with Gemini 3 Pro establishing the current peak at 38%).*

Standard Run Configurations

- **Setup:** Zero-shot, CoT allowed, Temperature 0, Pass@1, absolutely no external tools or internet search.
- **Language:** Strictly English-only.

Criticisms & Known Issues of HLE

- **Factual/Label Disputes:** Because the questions push the absolute limits of human knowledge, there are severe disputes over what constitutes a "correct" answer. Originally a 3,000-question dataset, **500 questions had to be completely deleted** due to expert disputes, reducing the active set to 2,500.
- **LLM-as-a-Judge Errors:** Grading 80% short-answer questions requires an LLM-as-a-judge, which introduces grading errors and inconsistencies.
- **Selection Bias:** Curated specifically from questions that stumped 2024 frontier models, creating a localized selection bias rather than a natural distribution of academic knowledge.

12. Summary Matrix: The Evolution of Knowledge Benchmarks

Benchmark	Launch Date	Format	Size	Core Target	Status (2026)
MMLU 7	Sept 2020 7	4-Option MCQ 11	~14,000 Qs 7	Factual Breadth (57 subjects) 7 19	Saturated & Retired 15
TruthfulQA 24	Sept 2021 28	MC2 (Default) & Gen 32 33	817 Qs 24	Common Human Misconceptions 24	Saturated & Retired 30 39
AGI Eval 48	Apr 2023 49	MCQ & Short Ans 52	~8,000 Qs 52	Standardized Human Exams 49	Saturated & Retired 54
GPQA 26	Nov 2023 56	4-Option MCQ	198 Qs	PhD-level	Nearing

Benchmark	Launch Date	Format	Size	Core Target	Status (2026)
		59	(Diamond) 60	Science Depth 26 58	Saturation 56
MMLU Pro 26	2024 70	10-Option MCQ 11	~12,000 Qs 69	Reasoning- focused MCQ Breadth 11 67	Nearing Saturation 70
SimpleQA 27	Late 2024 40	Short-Answer 40	4,326 Qs 40	Short-form Factual Recall & Calibration 27 42	Active 41 44
HLE 73	Jan 2025 74	80% Short- Ans / 20% MCQ 76 79	2,500 Qs 73	Depth × Breadth PhD- level (Multimodal) 75 ...	Active (SOTA) 74 77